

Housing, Fertility, and Demographic Transition

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Research question

How do housing costs and access to family-sized space affect fertility when births also reshape future population and housing demand?

- Children raise current demand for rooms when households have little liquid wealth.
- Tenure and down-payment constraints determine which households can obtain those rooms.
- Birth cohorts later become persons and household heads, feeding back into housing demand.

Takeaway. Housing is both a current input into family formation and a market whose future scale depends on family formation.

Family size and housing demand

Data object	Estimate	Interpretation
Four-year rooms contrast at first birth	+0.720	Housing expands around entry into parenthood
Rooms gap: 3+ versus 1–2 children	+0.368	Larger families occupy more space
Family ownership gap	+16.8 pp	Families are more likely to own

- The first-birth contrast is a PSID event-time -1 to $+3$ comparison, not a causal estimate.
- The other two moments are cross-sectional ACS targets in the active calibration.

Takeaway. The same family transition that raises demand for rooms also moves households toward ownership.

Source: Project target ledger: PSID, ACS/IPUMS, and the active 2023 calibration contract.

Access to family-sized housing

Descriptive object	Share	Economic margin
Six-plus-room owner stock held by heads age 46+	71.8%	Intergenerational allocation
Six-plus-room owner stock held by households without children at home	54.9%	Retention after children leave
Largest rental unit in the quantitative menu	6 rooms	Upper tail of the rental-size distribution
Required buyer equity in the benchmark	20%	Liquid-wealth access

The friction is not that older households occupy housing. It is that the marginal value of family space can be high for young constrained households while much of the large owner stock is held elsewhere.

Source: ACS/IPUMS matched-metro tabulations in the circulated model note; active housing menu and financing restriction.

Contribution

1. **Household mechanism:** sequential fertility, housing size, tenure, saving, income risk, and warm-glow bequests are chosen jointly.
2. **Demographic equilibrium:** births create persons; survival, migration, and head formation create the household-head stock.
3. **Transition equilibrium:** forward-looking asset prices, housing supply, and an equal property-tax rebate clear jointly with household and demographic dynamics.

The active implementation is a single national housing market. Location is therefore degenerate here; spatial sorting is a separate extension.

Takeaway. Separating persons from household heads changes both the transition law and the definition of a steady state.

The full cost of a child

In a compact model, a young household chooses adult housing h and children n . One child requires a units of space and q units of other goods. Limited access to family housing adds a goods-equivalent wedge z :

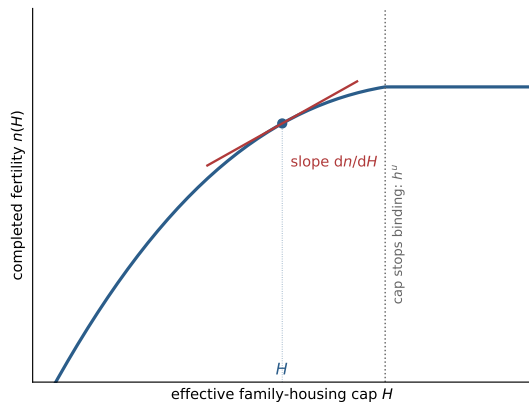
$$n(r, z) = \frac{v}{q + a(r + z)}, \quad d_Y(r, z) = \frac{\alpha}{r} + a n(r, z).$$

- r is the current price of housing services.
- z summarizes the rental-size limit and the down-payment constraint.
- A higher r or z raises the full resource cost $q + a(r + z)$ of a child.

Takeaway. Housing access acts like a household-specific addition to the price of children.

Source: Compact analytical model; the representative-family law is intuition, not the quantitative demographic closure.

The effective housing cap



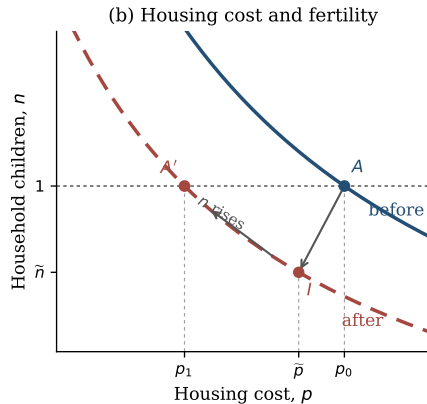
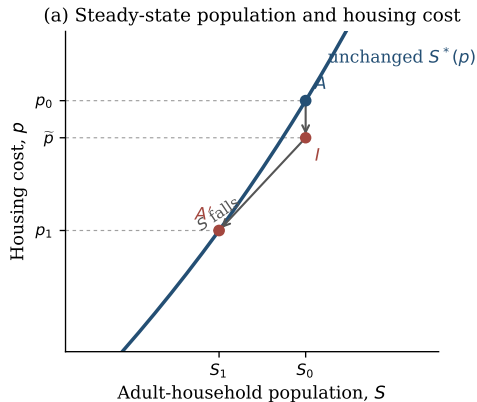
Source: Analytical model figure; P is the housing asset price and ϕ is the financed share.

The accessible family-space cap is

$$H_i^R = \bar{h}^R, \quad H_i^O = \min \left\{ \bar{h}^O, \frac{b_i}{(1 - \phi)P} \right\}.$$

- Renters face the largest available rental unit.
- Buyers need cash for the down payment $(1 - \phi)PH$.
- Relaxing a binding cap raises fertility along $n(H)$.

Housing–fertility feedback



In the closed toy economy, reproductive stationarity selects the housing cost and housing clearing selects population scale. The lifecycle model replaces this representative-family condition with the full person-cohort and household-head fixed point.

Source: Analytical transition note; the figure illustrates the mechanism, not the calibrated U.S. closure.

Lifecycle timing and state

One period is four years. A household enters at age 18, can live through age 85, and carries the beginning-of-period state

$$x = (b, z, a, o, n, \mathbf{c}).$$

- b : liquid net wealth; z : persistent earnings; a : age.
- o : renter or owner product; n : literal parity; $\mathbf{c}$: independent dependent-child clocks.
- Fertility is sequential: a successful birth raises parity by one and starts a child clock.
- The active national specification has one location; no migration choice is assigned to households.

Takeaway. Children affect current housing needs immediately, but their effect on the number of household heads arrives through a separate demographic law.

Preferences and child space

Let m be the number of dependent children currently at home. Flow utility is

$$u_t(c, s; m) = e(m)^{\sigma-1} \frac{\left[c^{\alpha_0} (s - \bar{h}(m))^{1-\alpha_0} \right]^{1-\sigma}}{1-\sigma} + \psi_t m,$$

with

$$e(m) = \left(\frac{2 + 0.7m}{2} \right)^{0.7}, \quad \bar{h}(m) = h_1 \mathbf{1}\{m > 0\} + h_m m.$$

- $e(m)$ is a consumption equivalence scale.
- h_1 is a first-child room jump and h_m is the per-child room floor.
- Owner services are $s = \chi H_k$; renter services are the chosen rental rooms.

Housing, finance, and bequests

- Renters choose continuous rooms up to 6; owners choose $H_k \in \{2, 4, 6, 8, 10\}$.
- A purchase requires cash of at least $(1 - \phi)P_t H_k$, with $\phi = 0.80$.
- Owners pay depreciation and property tax and incur a proportional moving cost when changing products.
- Liquid wealth earns a risk-free return and can be negative only within the collateral and age-tapered debt rules.

Warm-glow bequest utility is

$$\mathcal{B}(w^e) = \theta_0 \frac{(\theta_1 + \max\{w^e, 0\})^{1-\sigma}}{1-\sigma}.$$

Estates affect saving and old-age housing retention. They are not assigned to particular descendants; entrant wealth is externally distributed.

Sequential fertility and the Bellman problem

At an eligible state, the household compares waiting with attempting the next birth. Let $d = 1$ denote an attempt, let π_a be age- a conception success, and let x^+ raise parity by one and start a child clock. Then

$$\Pr(d = 1 \mid x, t) = \Lambda\left(\frac{\pi_a[\mathcal{V}_t(x^+) - \mathcal{V}_t(x) - F_1 \mathbf{1}\{n = 0\}]}{\kappa(n)}\right),$$

where $\mathcal{V}_t$ already optimizes tenure, rooms, consumption, and saving conditional on the parity branch. A birth occurs with probability $\pi_a \Pr(d = 1 \mid x, t)$; $\kappa(0) = \kappa_1$ and $\kappa(n \geq 1) = \kappa_C$. Childless households must also satisfy the readiness gate before attempting.

Takeaway. Housing affects fertility through current child-space needs, tenure feasibility, and the continuation value of wealth and housing.

Housing-market equilibrium

Household policies imply occupied housing demand D_t . Supply is static elastic around the 2023 anchor:

$$H_t^s = H_{2023} \left(\frac{P_t}{P_{2023}} \right)^\eta, \quad \eta = 1.75, \quad D_t = H_t^s.$$

Asset prices and rents satisfy the one-period no-arbitrage identity

$$r_t + P_{t+1} = (R + \delta + \tau_H)P_t.$$

- P_t capitalizes the next price; r_t is the current renter price.
- R is the four-year gross liquid return; δ and τ_H are four-year wedges.
- Households solve backward along the price and transfer paths.
- The distribution, births, persons, and heads move forward.

From the estimation bridge to the transition law

Dates	Population object	Role
2007–2023	Census household totals and ACS head-age shares	External bridge used to evaluate a dated calibration
2023	Persons, heads, and the household state distribution	Joint initial condition
After 2023	Births, person cohorts, survival, migration, and head formation	Endogenous transition accounting

The pre-2023 bridge is an imposed historical input, not a model fit. It is not extrapolated forward. Post-2023 births are never converted mechanically into future heads by dividing by 2.1.

Takeaway. The estimated household model and the forward demographic closure meet at the reconstructed 2023 state.

Persons are not household heads

2023 stock	Level	Accounting role
Resident persons	338.7 million	Cohorts exposed to survival and migration
Household heads	129.0 million	Economic decision makers in the lifecycle model
Persons per head	2.63	Endogenous ratio of two distinct stocks

Headship is low at young adult ages and rises sharply as nonheads form households. A birth changes the person stock now; it can change the head stock only later through survival and household formation.

Source: Accepted 1% transition initial state and the aligned 2023 ACS age–sex headship schedule.

The person-cohort law

Let $N_t^{g,a}$ denote resident persons of sex g and single year of age a . The annual law is

$$N_{t+1} = A_{t+1}N_t + f_{t+1}B_{t+1} + M_{t+1},$$

where

- A_{t+1} ages existing cohorts using destination-age survival;
- $f_{t+1}B_{t+1}$ allocates newborn survivors by sex at age zero;
- M_{t+1} is the age–sex net-migration flow after survival.

The model produces a four-year birth flow. It is spread uniformly across the four calendar years, while each year uses its own sex shares, survival, and migration inputs.

Takeaway. Births create persons; survival and migration determine which cohorts are resident four years later.

Head formation and the household distribution

The head stock is a separate accounting object. In each age–sex cell, surviving heads are adjusted by the minimum one-way flow needed to attain the externally fixed headship rate $\rho^{g,a}$:

$$H_{t+1}^{g,a} = \rho^{g,a} N_{t+1}^{g,a} = H_{t+1}^{\text{surv},g,a} + F_{t+1}^{g,a} - D_{t+1}^{g,a} + H_{t+1}^{M,g,a}.$$

The economic household operator supplies a raw within-age distribution $\tilde{G}_{t+1,a}$. Its mass is rescaled to the person-law head target:

$$G_{t+1,a}(x) = H_{t+1,a} \frac{\tilde{G}_{t+1,a}(x)}{\int \tilde{G}_{t+1,a}(x') dx'}.$$

The conditional wealth, tenure, income, parity, and child-state composition is preserved within age; this is an accounting closure, not an estimated household-formation hazard.

Transition equilibrium

A perfect-foresight transition consists of paths

$$\{V_t, \pi_t, G_t, N_t, H_t, P_t, r_t, T_t\}_{t=2023}^T$$

such that, at every date:

1. household policies π_t solve the Bellman problem given future prices and transfers;
2. births from G_t advance persons and heads, and the household distribution is rebalanced to head mass;
3. occupied housing equals static-elastic supply;
4. the asset-price/rent identity holds;
5. property-tax revenue equals the equal transfer paid to household heads.

Takeaway. Prices are forward looking, while the person and household distributions move forward from the same initial condition.

The corrected steady state

A steady state is a constant tuple

$$(N^*, H^*, G^*, \pi^*, P^*, r^*, T^*)$$

satisfying all of the following simultaneously:

- household optimality and household-distribution stationarity;
- the person-cohort fixed point under the stated survival and migration closure;
- head formation and exact agreement between head mass and household mass;
- housing-market clearing and the stationary asset-price/rent identity;
- the government budget and demographic reproduction.

A finite path is not a steady state. Nor is a constant household distribution whose implied births fail to reproduce the person and head stocks.

Externally fixed terminal objects

Object	Terminal treatment	Classification
Survival	2100 age–sex schedule held fixed	External
Net migration	2100 inferred age–sex flow held fixed in levels	External
Birth sex shares	2100 shares held fixed	External
Headship	Aligned 2023 ACS age–sex rates held fixed	Empirical accounting normalization
Fertility preference	$\psi = -0.069659$	Fitted 2023 value
Housing supply elasticity	$\eta = 1.75$	External benchmark
Property tax and rebate	Policy-specific, equal per head	Counterfactual contract

Takeaway. The terminal equilibrium is conditional on fixed migration, survival, headship, and policy objects; it is not an unconditional national forecast.

Dated calibration

The model is estimated at a point along the 2007–2023 transition.

1. The 2007 fertility intercept reproduces completed fertility of 2.1 in a constructed old benchmark.
2. Only the age marginal is reweighted to the 2007 ACS head distribution.
3. A reduced-form fertility-preference shifter moves linearly through 2023.
4. Census household totals and ACS head-age shares are imposed at five dates.
5. Twelve moments, principally measured on the 2023 state, discipline eleven free parameters.

The preference path is not interpreted as a causally identified shock date, and the imposed household bridge is not counted as fit.

Fertility, timing, and family housing fit

Moment	Target	Model
Completed fertility, women 40–44	1.918	1.918
Childless share, women 40–44	0.188	0.186
Mean age at first birth	26.045	26.327
First births at age 30 or later	0.260	0.242
Four-year rooms contrast at first birth	0.720	0.380
Rooms gap, 3+ versus 1–2 children	0.368	0.359

Takeaway. Fertility stocks fit closely; first-birth timing is reasonable; the model produces only about half of the first-birth rooms contrast.

Source: Active target set e5_fullhistory_roomsfix_h1_20260817.

Tenure, rooms, and wealth fit

Moment	Target	Model
Family ownership gap	0.168	0.177
Ownership rate, ages 30–55	0.575	0.546
Mean occupied rooms, ages 18–85	5.780	6.710
Wealth / annual gross labor earnings	6.873	6.766
Annual bequest flow / aggregate wealth	0.0088	0.0086
Old wealth-to-income p90/p50	3.448	3.396

Takeaway. Tenure and wealth are close, but the model overstates the level of occupied rooms.

Source: Active 2023 target-fit packet; complete weighted-loss accounting is in the appendix.

Remaining fit and identification

The strict SMM loss is 36.099. Two housing moments account for 26.304, or 73% of the objective:

- first-birth rooms contrast: model 0.380 versus target 0.720;
- mean occupied rooms: model 6.710 versus target 5.780.

The weighted Jacobian has rank 10/11 at the predeclared relative singular-value threshold 10^{-3} and condition number 1,104. Its weakest direction loads primarily on the bequest shift θ_1 .

Takeaway. The estimate is reproducible and interior, but locally weakly identified and not a claim of global optimality.

Source: Final coordinate identification audit around the selected parameter vector.

The accepted transition reference

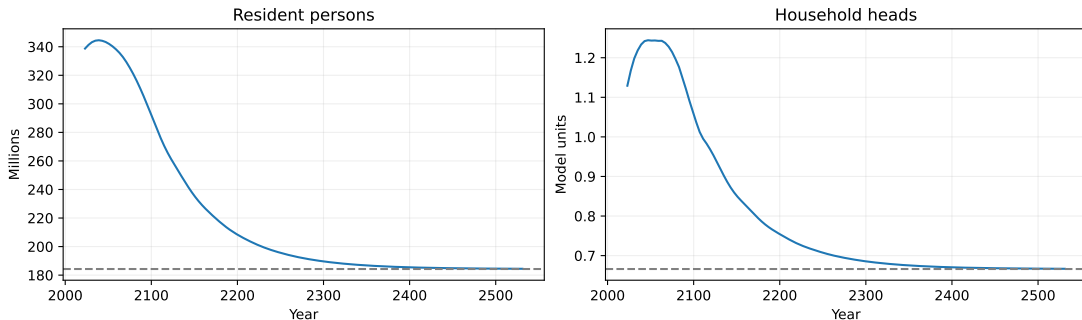
Object	Annual tax	Rebate	Role
Calibration status quo	1%	None	Preserved fitted 2023 reference
Transition reference	1%	Equal per head	Accepted 2023-forward equilibrium path
Policy reform	2%	Equal per head	Permanent tax change under the same closure

The transition reference begins from the fitted 2023 household state and the reconstructed 2023 person/head stocks. Supply is reanchored so the rebated 1% economy clears at the inherited quantity, and the terminal fertility preference remains at its fitted 2023 value.

Takeaway. The accepted transition baseline is rebated; it is not the unreputed calibration status quo.

Demographic momentum

Rebated 1% property tax

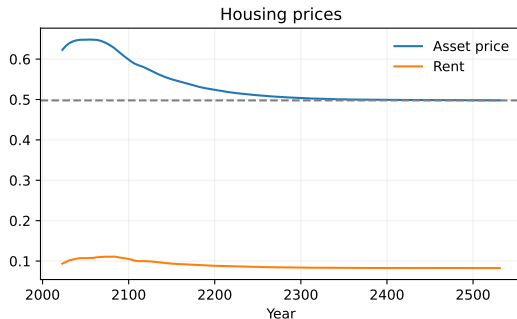
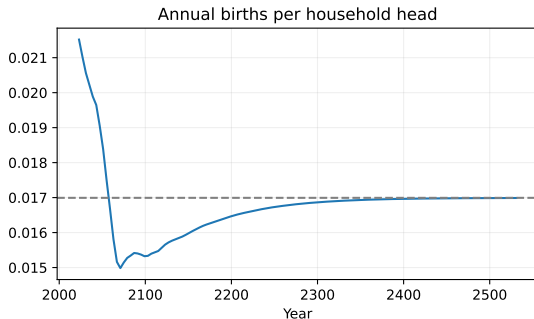


Resident persons rise from 338.7 million in 2023 to 344.6 million in 2039; household heads peak later, at 142.1 million in 2047. Both then contract toward the conditional terminal stocks.

Takeaway. An inherited age distribution can generate decades of momentum even when subsequent births are low.

Source: Accepted H128 rebated 1% transition. Dashed lines are its jointly solved conditional terminal equilibrium.

Prices and births on different clocks



The asset price peaks in 2055; the renter price peaks in 2083 because it also reflects the next-period asset payoff. Annual births per head undershoot the terminal rate before converging back toward it.

Takeaway. Cohort composition, forward-looking capitalization, and household fertility need not turn at the same date.

Source: Accepted H128 rebated 1% transition; four-year model births are divided by four in the plotted annual rate.

The conditional terminal equilibrium

Object	2023	Terminal	Change
Resident persons, millions	338.724	184.278	−45.6%
Household heads, millions	128.964	76.073	−41.0%
Asset price	0.6229	0.4977	−20.1%
Rent-equivalent price	0.0936	0.0825	−11.8%
Annual births per head	0.02152	0.01699	−21.0%

The terminal demographic renewal ratio is 0.5568. A constant population is possible because the fixed age–sex migration flow supplies the remaining person cohorts; without that flow, this is not a positive closed steady state.

Takeaway. The endpoint is a conditional fixed-migration equilibrium, not a forecast of the U.S. population level.

The rebated property tax

The reform permanently raises the annual property-tax rate from 1% to 2% and returns all contemporaneous revenue equally across household heads. With $\tau_{H,t} = 4\tau_{H,t}^{\text{ann}}$ in a four-year model period,

$$T_t \int dG_t = \tau_{H,t} P_t \int H(o_t) dG_t.$$

- **Capitalization:** the higher carrying cost tends to reduce the asset price.
- **User cost:** the renter price depends on both the tax rate and the next asset price.
- **Tenure and fertility:** down payments, owner products, and child-space choices adjust state by state.
- **Demographic scale:** changed births affect persons immediately and heads only with a lag.

This is an equilibrium incidence exercise, not a welfare comparison.

Source: Policy contract: identical initial state, survival, migration, headship, supply elasticity, and terminal fertility preference across tax cases.

Conclusion

1. Children raise demand for rooms when rental size and down payments constrain access to family housing.
2. A lifecycle model is needed because fertility, tenure, saving, and old-age housing retention are jointly determined.
3. A coherent transition must move persons first and derive household heads through survival, migration, and formation.
4. The fitted model matches fertility stocks, tenure, and wealth more closely than housing quantities and is locally weakly identified.
5. The accepted rebated 1% path displays demographic momentum followed by a slow transition to a conditional fixed-migration equilibrium.

Takeaway. Housing policy can affect family formation today and the scale of housing demand generations later; the two margins require one internally consistent equilibrium law.

Appendix

Compact household derivation

The young household in the simple model solves

$$\max_{c,h,n} c + \alpha \log h + v \log n \quad \text{s.t.} \quad c + rh + [q + a(r + z)]n = y.$$

Interior first-order conditions give

$$h(r) = \frac{\alpha}{r}, \quad n(r, z) = \frac{v}{q + a(r + z)}, \quad d_Y(r, z) = \frac{\alpha}{r} + an(r, z).$$

Thus

$$\frac{\partial n}{\partial r} = \frac{\partial n}{\partial z} = -\frac{av}{[q + a(r + z)]^2} < 0.$$

The quantitative model preserves the cost channel but replaces continuous fertility, space, and tenure with sequential fertility, a rental cap, and a discrete owner ladder.

Access-gap interpretation

For a constrained renter, divide the housing first-order condition by the marginal utility of consumption:

$$\frac{u_h}{u_c} = r + \zeta_i^R, \quad \zeta_i^R \geq 0.$$

For an owner, the same gap separates the down-payment and size constraints:

$$\zeta_i^O = \underbrace{\frac{\mu_i}{\lambda_i}(1 - \phi)P}_{\text{financing}} + \underbrace{\frac{\theta_i}{\lambda_i}}_{\text{owner size limit}}.$$

If each child requires a units of space, $a\zeta_i$ is the shadow child cost generated by limited access. The gap is a local marginal valuation, not the welfare gain from an entire extra room.

Bellman timing

Within a four-year date, the household:

1. observes fertility taste shocks and, when eligible, chooses whether to wait or attempt a birth;
2. realizes conception success and chooses renter rooms or an owner product conditional on parity;
3. chooses consumption and next-period liquid wealth;
4. receives income and pays rent, property tax, depreciation, and any moving cost;
5. ages children independently, transitions income, and survives or leaves a warm-glow estate.

In a transition, date- t policies use the date- $(t + 1)$ continuation value. The terminal value function and terminal next-date asset price are boundary conditions; they do not replace the requirement that the simulated terminal distribution approach the terminal fixed point.

Exact annual person accounting

The annual law can be written cell by cell. For newborns,

$$N_{t+1}^{g,0} = s_{t+1}^{g,0} f_{t+1}^g B_{t+1} + M_{t+1}^{g,0}.$$

For interior ages $a \geq 1$,

$$N_{t+1}^{g,a} = s_{t+1}^{g,a} N_t^{g,a-1} + M_{t+1}^{g,a}.$$

The terminal age is top-coded, so survivors from the penultimate and terminal cells both enter the last cell. Every step satisfies

$$\sum N_{t+1} = \sum N_t + B_{t+1} - \text{newborn deaths} - \text{existing deaths} + \sum M_{t+1}.$$

Exact household-head accounting

Surviving heads are aged with the same survival schedule. In each cell, the target is $\rho^{g,a} N_{t+1}^{g,a}$. The accounting closure uses only one gross adjustment direction:

$$F_{t+1}^{g,a} = \max\{\rho^{g,a} N_{t+1}^{\text{preM},g,a} - H_{t+1}^{\text{surv},g,a}, 0\},$$

$$D_{t+1}^{g,a} = \max\{H_{t+1}^{\text{surv},g,a} - \rho^{g,a} N_{t+1}^{\text{preM},g,a}, 0\}, \quad H_{t+1}^{M,g,a} = \rho^{g,a} M_{t+1}^{g,a}.$$

Hence $H_{t+1}^{g,a} = \rho^{g,a} N_{t+1}^{g,a}$ exactly. Formation and dissolution are accounting flows, not simultaneous estimated hazards.

The within-age household bridge

Let $\tilde{G}_{t+1,a}(x)$ be the raw household transition within age a , and let $H_{t+1,a}$ be the head mass implied by persons. Then

$$G_{t+1,a}(x) = \begin{cases} H_{t+1,a} \frac{\tilde{G}_{t+1,a}(x)}{\int \tilde{G}_{t+1,a}(x') dx'}, & \int \tilde{G}_{t+1,a} > 0, \\ H_{t+1,a} \Gamma_a(x), & \text{for an empty entrant age cell,} \end{cases}$$

where Γ_a is the externally specified entrant template. The operation exposes added and removed mass separately and moves no mass across ages.

A richer model would estimate economic states for newly formed, dissolved, and migrant heads rather than preserve the raw conditional composition.

Steady-state demographic scale

With fixed terminal survival A , newborn-survivor vector f , net migration m , and headship vector ρ , persons satisfy

$$N^* = AN^* + f b^*(\rho' N^*) + m,$$

where b^* is annual births per head implied by the household policy.

Define the person responses

$$v_B = (I - A)^{-1}f, \quad v_M = (I - A)^{-1}m.$$

Then the demographic renewal ratio is

$$\mathcal{R} = b^* \rho' v_B,$$

and, when fixed migration is positive, a finite steady state requires $\mathcal{R} < 1$. Annual births and persons are

$$B^* = \frac{b^* \rho' v_M}{1 - \mathcal{R}}, \quad N^* = v_M + B^* v_B.$$

Market and fiscal closure

At every date, occupied physical rooms satisfy

$$\int h(x; \pi_t) dG_t(x) = H_{2023} \left(\frac{P_t}{P_{2023}} \right)^\eta.$$

The rebated tax satisfies

$$\tau_{H,t} P_t \int H(o_t) dG_t = T_t \int dG_t.$$

At a stationary price,

$$r^* = (R - 1 + \delta + \tau_H) P^*.$$

Housing demand uses physical rooms. The owner-service premium affects utility but does not create additional physical supply. Conditional renter housing is also distinct from realized housing after the tenure choice.

Complete target fit I

Moment	Target	Model	Gap	Weight	Loss
Completed fertility, women 40–44	1.918000	1.918251	0.000251	1,425.739	0.000090
Childless share, women 40–44	0.188000	0.185867	−0.002133	17,180.744	0.078133
Mean age at first birth	26.044627	26.327334	0.282707	44.444	3.552148
First births at age 30 or later	0.260327	0.242115	−0.018213	10,000.000	3.316971
Four-year rooms contrast at first birth	0.720246	0.379786	−0.340461	137.565	15.945664
Rooms gap, 3+ versus 1–2 children	0.367700	0.359253	−0.008447	2,958.515	0.211071

Completed fertility and childlessness are CPS age-42 cohort stocks. First-birth timing uses a fixed synthetic cohort. The rooms response follows the 2019 childless risk set through matched birth/no-birth branches to 2023.

Complete target fit II

Moment	Target	Model	Gap	Weight	Loss
Family ownership gap	0.167662	0.176866	0.009204	14,229.591	1.205562
Ownership rate, ages 30–55	0.575472	0.546060	−0.029412	1,207.846	1.044837
Mean occupied rooms, ages 18–85	5.779970	6.710105	0.930135	11.973	10.358582
Wealth / annual gross labor earnings	6.873100	6.766352	−0.106748	6.288	0.071648
Annual bequest flow / aggregate wealth	0.008800	0.008623	−0.000177	5,165,289.256	0.161538
Old wealth-to-income p90/p50	3.448111	3.396287	−0.051824	56.960	0.152979
Total strict loss					36.099223

All twelve moments enter the same active objective. The first-birth rooms row is retained as a descriptive contrast despite its non-flat prepath.

Free parameters I

Parameter	Economic interpretation	Estimate	Search bound
β_{annual}	Patience	0.994959	[0.9400, 0.9995]
κ_1	First-birth logit scale	2.267631	[0.02, 50]
κ_C	Continuation-birth logit scale	1.765427	[0.02, 50]
χ	Owner-service premium	1.033945	[0.10, 5]
H_0	Housing-supply scale	16.381189	[0.20, 80]
θ_0	Bequest scale	0.563219	[0, 8]

No estimate is within two percent of its transformed search boundary.

Free parameters II

Parameter	Economic interpretation	Estimate	Search bound
θ_1	Bequest wealth shift	0.099707	[0.02, 16]
h_m	Per-child room floor	0.315267	[0.10, 1.80]
h_1	First-child room jump	0.199446	[0, 0.50]
F_1	First-birth fixed cost	4.545167	[0, 8]
$\Delta\psi_{2007:2023}$	Fertility-preference change	-0.350000	[-1.50, 0.20]

The implied fertility intercepts are $\psi_{2007} = 0.280341$ and $\psi_{2023} = -0.069659$. The intercept change is a reduced-form residual, not a causally identified preference shock.

Assigned household and market objects

Parameter	Value	Source or restriction
Model period	4 years	Timing
Entry and maximum ages	18 and 85	Timing
Risk aversion σ	2	Maintained
Financed share ϕ	0.80	20% buyer-equity restriction
Sale cost	6%	Housing literature benchmark
Owner room menu	{2, 4, 6, 8, 10}	ACS occupied-room distribution
Rental cap	6 rooms	ACS rental-size distribution
Supply elasticity η	1.75	Saiz benchmark
Entrant wealth	PSID ratio distribution	Childless renters age 18–24
Tenure taste-shock scale	0	Author restriction
Child dependence of bequests	0	Child-invariant warm glow

Historical calibration bridge

The 2007–2023 calibration uses five four-year dates. At each date, the observed household total and head-age marginal are imposed before the twelve maintained moments are evaluated.

- This absorbs household formation and migration that births since 2007 cannot generate by 2023.
- It makes the cross-section a dated state rather than a stationary distribution.
- Historical fertility, ownership, rooms, and price paths are holdouts unless explicitly included in the target system.
- The bridge is retired after 2023 and replaced by the person/head law.

Consequently, matching the imposed population path is an accounting identity, not evidence that the model explains U.S. household growth over 2007–2023.

Local identification audit

At the selected vector, the standardized residual is

$$r_j(\theta) = \sqrt{w_j} [m_j(\theta) - \hat{m}_j].$$

A fresh two-sided coordinate panel constructs the weighted Jacobian $J = \partial r / \partial \theta$.

Jacobian	Rank at 10^{-3}	Condition number
All 11 free parameters	10/11	1,104.01
Ten active columns after freezing the inconsistent bequest column	10/10	391.02

The weakest right-singular vector places 91.9% of its absolute loading on θ_1 . Local weak identification is therefore concentrated in the bequest block.

Accepted baseline equilibrium checks

Object	Maximum absolute residual	Interpretation
Housing market	1.7943×10^{-4}	Demand relative to supply
Government budget	1.1572×10^{-5}	Revenue minus equal transfers
Person identity	numerical precision	Births, deaths, and migration
Head identity	numerical precision	Deaths, formation, dissolution, migrant heads
Household/head identity	numerical precision	Household mass equals person-law head mass

The final endogenous date is also close to the independently solved terminal person distribution, head distribution, household distribution, price, rent, transfer, and fertility preference.

Slow terminal convergence

The dominant eigenvalue of the frozen-terminal four-year demographic block is

$$\lambda_1 = 0.960754.$$

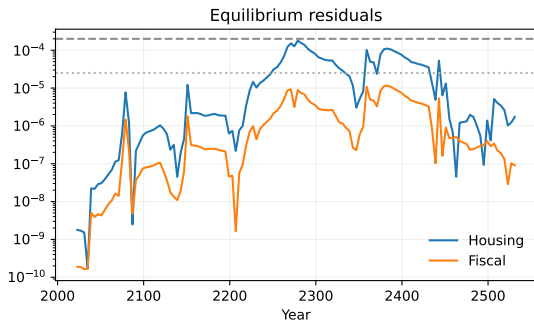
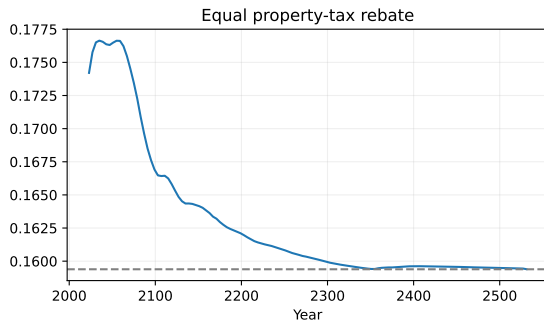
It implies

$$\text{half-life} = \frac{4 \log(1/2)}{\log \lambda_1} \approx 69 \text{ years}, \quad \lambda_1^{116} < 0.01.$$

Thus one percent attenuation of the slow mode requires about 116 model periods, or 464 years. The accepted path uses 128 four-year periods so the simulated state can approach the terminal fixed point rather than merely inherit it as a boundary condition.

Takeaway. The long horizon reflects age-structured demographic persistence, not a forecast horizon.

Baseline transition diagnostics



The equal rebate and equilibrium residuals are solved jointly with prices. The displayed residuals remain below their common acceptance thresholds throughout the path.

Source: Accepted H128 rebated 1% transition packet.

Policy comparison contract

A quantitative 2%–minus–1% comparison must hold fixed:

- the fitted 2023 household state and reconstructed person/head stocks;
- survival, net migration, birth sex shares, and headship;
- the housing-supply elasticity and 2023 supply anchor;
- the terminal fertility preference and all calibrated household parameters;
- the equal-rebate rule and the definitions of housing and fiscal residuals.

The comparison then separates the immediate response of prices, rents, tenure, and births from the delayed response of resident persons and household heads. No migration response or welfare ranking is part of this contract.

Interpretive boundaries

- The first-birth rooms contrast is descriptive and has a non-flat prepath.
- The active calibration is locally weakly identified in the bequest block.
- Fixed 2023 headship is an accounting closure, not a structural formation model.
- Fixed 2100 migration in levels is essential to the positive terminal population scale.
- Static-elastic supply omits durable construction and asymmetric decline dynamics.
- One national market omits spatial sorting and local migration.
- The policy exercise reports equilibrium incidence, not welfare.

These limitations narrow the interpretation without changing the exact accounting distinction between persons, heads, and household economic states.

Selected literature

Margin	Selected references
Lifecycle fertility	Becker and Lewis; Sommer; Doepke and Kindermann
Housing and childbearing	Lovenheim and Mumford; Dettling and Kearney; Hacamo
Housing supply and capitalization	Rosen; Roback; Saiz; Gyourko, Mayer, and Sinai
Housing tenure and user cost	Henderson and Ioannides; Sommer, Sullivan, and Verbrugge
Demography and housing demand	Mankiw and Weil; Francke et al.; Coeurdacier et al.
Spatial scale	Hsieh and Moretti; Rosen–Roback spatial equilibrium

The contribution here is to combine the lifecycle housing–fertility mechanism with an explicit person-to-head demographic transition and a jointly cleared housing and fiscal equilibrium.